SECURE POLLING BOOTH USING RFID AND LIVE VOTE TRACKING

A project submitted to the Bharathidasan University

in partial fulfillment of the requirements

for the award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted by

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Register Number: 235113331

Under the guidance of

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DEPARTMENT OF COMPUTER APPLICATIONS

BISHOP HEBER COLLEGE (AUTONOMOUS)

"Nationally Reaccredited at A^{++} Grade with a CGPA of 3.69 out of 4 in NAAC IV Cycle"

Recognized by UGC as "College of Excellence"

Affiliated to Bharathidasan University

TIRUCHIRAPPALLI-620 017.

OCTOBER – 2025

DECLARATION

I hereby declare that the project work presented is originally done by me under the guidance of **Mr. G.PAUL DAVIDSON MCA., M.Phil., Assistant Professor, UG Department of Computer Applications, Bishop Heber College (Autonomous), Tiruchirappalli-620 017** and has not been included in any other thesis/project submitted for any other degree.

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This is to certify that the project work entitled **“SECURED POLLING BOOTH USING RFID AND LIVE VOTE TRACKING ”** is a bonafide record work done by **SANTHIYA S, Register Number: 235113331** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF COMPUTER APPLICATIONS** during the period **2023 - 2026.**

Place: **Signature of the Guide**

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Date:

Course Title: Project **Course Code: U23CA5PJ**

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1.

2.

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SANTHIYA S

ABSTRACT

Traditional voting systems, which rely on paper ballots and manual voter verification, suffer from several disadvantages. The process is time-consuming, as long queues form due to manual checking and ballot casting. It is also prone to errors during vote counting, which may result in miscalculated or disputed results. Security issues such as impersonation, fake voting, and multiple voting attempts by the same person are common due to weak identity verification methods. Free and fair elections are the backbone of democracy, but the existing voting process often faces issues such as impersonation, slow counting, and limited transparency.

To overcome these challenges, this project proposes a secure electronic voting system using RFID authentication, password verification, and IoT-based monitoring. Each voter is identified by an RFID card, verified through a password, and then allowed to cast a vote via candidate-specific buttons. The chosen candidate is displayed on an LCD, while a GSM module sends an SMS confirming the voter's choice. Voting data is also updated on an IoT platform for real-time monitoring, ensuring accuracy, security, and transparency.

The system is designed with multiple layers of security, including encryption and controlled access, to prevent data tampering during storage or transmission. Minimizing manual intervention it reduces human error, speeds up the counting process, and ensures that results are both accurate and timely. Real-time insights also discourage malpractices and provide transparency, thereby increasing voter confidence in the electoral process.

The model prioritizes **security, transparency, and efficiency**. By reducing human errors and manual work, it speeds up the entire election process and makes results more reliable. Real-time tracking also prevents malpractices and builds greater trust among voters. In short, the system shows how combining **RFID technology with secure digital voting and live monitoring** can modernize elections, making them more secure, transparent, and trustworthy

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1.INTRODUCTION

Voting plays a vital role in ensuring the democratic rights of citizens by allowing them to elect representatives in a free and fair manner. The integrity of the voting system directly impacts the credibility of the election process. However, traditional voting systems such as paper ballots and manual verification methods suffer from several drawbacks. These include long queues due to slow verification, impersonation and duplicate voting, human errors in ballot counting, and delays in result declaration. Such limitations often raise doubts about the transparency and security of elections, highlighting the urgent need for a reliable and modernized voting system.

With advancements in embedded systems and communication technologies, electronic voting machines (EVMs) have gained attention for their ability to improve accuracy and efficiency. Yet, even conventional EVMs lack multi- level security checks and real-time result monitoring, which makes them vulnerable to tampering and reduces voter confidence. To address these gaps, this project proposes an enhanced electronic voting system that integrates RFID- based voter authentication, password verification, GSM messaging, and IoT connectivity.

In the proposed system, each voter is assigned a unique RFID card that serves as their digital identity. When scanned, the system verifies the card against a database and displays the voter's name on an LCD screen. To strengthen security, an additional password verification step is included before granting access to the voting stage. Once authenticated, the voter can cast their vote by pressing a candidate-specific button. The selected candidate's name is displayed on the LCD, and the system automatically increments both the candidate's individual vote count and the total vote tally.

Beyond local processing, the system ensures transparency and communication through GSM and IoT technologies. A GSM module instantly sends an SMS notification confirming the voter's choice, ensuring accountability and confidence. Simultaneously, the IoT module uploads the updated vote counts to a cloud-based platform, enabling real-time monitoring of votes across multiple locations. Election authorities can track live results, identify irregularities, and ensure fairness throughout the process.

By combining RFID, password authentication, GSM, and IoT, this project delivers a robust and tamper-proof voting mechanism. It enhances security by preventing impersonation and unauthorized voting, ensures accuracy by eliminating human counting errors, reduces delays in result processing, and builds trust among voters through transparency and confirmation messages. Thus, the proposed system addresses the shortcomings of traditional voting methods and contributes to the development of a secure, efficient, and modern election process.

2. EXISTING & PROPOSED SYSTEM

2.1 EXISTING SYSTEM

The existing online voting systems primarily focus on enhancing convenience by allowing voters to cast their votes remotely. Most systems authenticate voters using a unique username and password, followed by face recognition to ensure that the voter casts the vote only once. Typically, the system uses machine learning algorithms, such as K- Nearest Neighbors (KNN), to verify the voter's identity through a webcam-captured image. Once verified, the voter can cast their vote, and the system updates election results in real-time on the website. Additionally, dynamic voice feedback guides the voter throughout the process, improving usability. While this system provides remote access, face recognition, and real-time updates, it still relies heavily on internet connectivity and centralized server verification. Moreover, it may lack multi-factor authentication, GSM-based confirmation, and IoT-enabled live monitoring, which are crucial for enhancing security, transparency, and trustworthiness in elections.

2.2 PROPOSED SYSTEM

The proposed system aims to develop a secure, transparent, and efficient electronic voting system by integrating RFID authentication, password verification, candidate selection, GSM communication, and IoT-based real-time monitoring. Each voter is issued a unique RFID card that serves as their digital identity. Upon scanning the card, the system verifies the voter's identity and displays their name on an LCD. To enhance security, the voter is required to enter a personal password; only upon correct verification can they proceed to the voting stage. Once authenticated, the voter can cast their vote by pressing a candidate-specific button. The system immediately displays the selected candidate on the LCD and updates the vote count for both the chosen candidate and the total votes. A GSM module sends an SMS confirmation to the voter, ensuring transparency and accountability. Additionally, the system is connected to an IoT platform, allowing real-time monitoring of individual candidate votes and overall voting statistics from a remote location.

3.HARDWARE & SOFTWARE REQUIREMENTS

3.1 HARDWARE REQUIREMENTS

3.1.1 ARDUINO MICROCONTROLLER

The Arduino Uno is a microcontroller board based on the ATmega328 (datasheet). It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Uno differs from all preceding boards in that it does not use the FTDI USB-to-serial driver chip. Instead, it features the Atmega16U2 (Atmega8U2 up to version R2) programmed as a USB-to-serial converter.

Fig. 3.1.1 ARDUINO MICROCONTROLLER

3.1.2 RFID Reader(RDM 6300) and RFID Cards

An RFID reader is a device that is used to interrogate an RFID tag. The reader has an antenna that emits radio waves; the tag responds by sending back its data. A number of factors can affect the distance at which a tag can be read (the read range). The frequency used for identification, the antenna gain, the orientation and polarization of the reader antenna and the transponder antenna, as well as the placement of the tag on the object to be identified will all have an impact on the RFID system's read range.

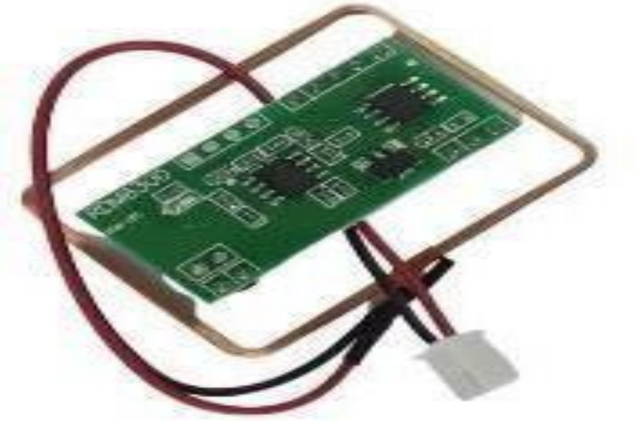

Fig. 3.1.2 RFID READER

An RFID tag is a microchip combined with an antenna in a compact package; the packaging is structured to allow the RFID tag to be attached to an object to be tracked. "RFID" stands for Radio Frequency Identification. The tag's antenna picks up signals from an RFID reader and scanner and then returns the signal, usually with some additional data (like a unique serial number or other customized information). RFID tags can be very small - the size of a large rice grain. Others may be the size of a small paperback book.

Fig. 3.1.2 RFID CARD

3.1.3 LCD Display (16x2)

A **16x2 LCD display** is one of the most commonly used modules in electronic projects for displaying text. The term *16x2* means it can show **16 characters per line** across **2 lines**, making a total of 32 characters at a time. It is based on liquid crystal display (LCD) technology and typically works using the **HD44780 controller**, which is an industry standard. Each character is displayed in a **5x8 dot matrix**, allowing clear and readable output. The module usually has **16 pins**, which include power supply pins, contrast adjustment, control pins (RS, RW, E), data pins (D0–D7), and sometimes backlight pins. It can be operated in both **8-bit mode** and **4-bit mode**, where 4-bit mode reduces the number of I/O pins required from the microcontroller.

The display is widely used with **Arduino, PIC, AVR, ARM, and Raspberry Pi** for applications such as digital clocks, calculators, embedded systems, and monitoring devices. Its advantages include **low cost, easy interfacing, and low power consumption**, while its limitation is that it can only display **alphanumeric characters and simple symbols**, not graphics. Despite this, the 16x2 LCD remains a reliable and popular choice for beginners and professionals in electronics.

Fig. 3.1.3 LCD DISPLAY

3.1.4 Keypad (3x4 Matrix)

A **keypad** is an input device commonly used in electronic systems to enter numbers, characters, or commands. It consists of a set of keys arranged in a matrix format, usually in **rows and columns**, which reduces the number of microcontroller pins required for connection. The most widely used type is the **4x4 keypad**, which has 16 keys arranged in 4 rows and 4 columns, though **4x3 (12-key)** keypads are also very popular for calculators, telephones, and access control systems. When a key is pressed, it establishes an electrical connection between a specific row and column, and the microcontroller detects the key's position by scanning the matrix. Keypads are inexpensive, durable, and simple to interface with microcontrollers such as Arduino, PIC, AVR, or ARM.

They are widely used in **ATMs, security systems, door locks, calculators, mobile devices, and embedded projects** where manual input is required. While they are limited in speed and can only handle one or a few key presses at a time, keypads remain one of the most effective and reliable input methods for small-scale electronic applications.

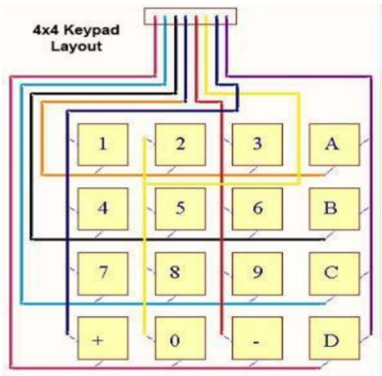

Fig. 3.1.4 KEYPAD (3x4 MATRIX)

3.1.5 Push Buttons

A push button switch is used to either close or open an electrical circuit depending on the application. Push button switches are used in various applications such as industrial equipment control handles, outdoor controls, mobile communication terminals, and medical equipment, and etc.

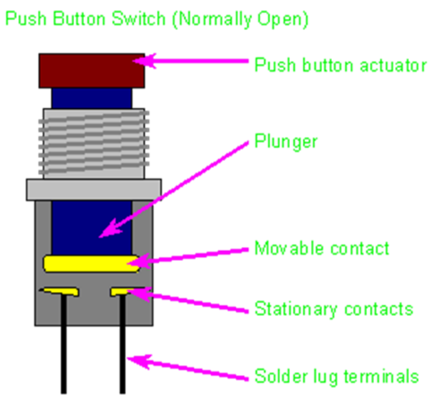

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Fig. 3.1.5 PUSH BUTTON

3.1.6 GSM Module (SIM800L)

GSM, which stands for *G*lobal *S*ystem for *M*obile communications, reigns as the world's most widely used cell phone technology. Cell phones use a cell phone service carrier's GSM network by searching for cell phone towers in the nearby area. GSM networks operate in a number of different carrier frequency ranges (separated into GSM frequency ranges for 2G and UMTS frequency bands for 3G), with most 2G GSM networks operating in the 900 MHz or 1800 MHz bands. Where these bands were already allocated, the 850 MHz and 1900 MHz bands were used instead (For example Canada and United States).

Fig. 3.1.6 GSM MODULE

3.1.7 Power Supply (5V)

A 5V power supply is one of the most essential components in electronic circuits and embedded systems for power. Many microcontrollers, sensors, displays, and modules operate at 5 volts DC, making a regulated supply necessary for stable operation. A typical 5V power supply is built using a step-down transformer, rectifier, filter, and voltage regulator. The transformer reduces AC mains voltage, which is then converted to DC by a rectifier. A capacitor filter smooths out the ripples, and finally, a voltage regulator such as 7805 provides a constant 5V output regardless of input fluctuations.switching power supplies and USB adapters are also commonly used to provide compact and efficient 5V outputs. It is widely used in Arduino boards,LCD modules, keypads, sensors, and communication devices. The main advantages of a 5V power supply are stability, low noise, and compatibility with a wide range of electronic components. Since most digital circuits require regulated power, the 5V power supply is considered the backbone of small electronic projects and embedded applications.

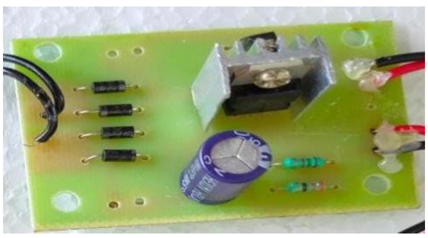

Fig. 3.1.7 POWER SUPPLY

Fig. 3.1.7 DC CONVERTOR

3.1.8 IoT (ESP8266)

The **ESP8266** is a low-cost Wi-Fi microchip with full TCP/IP capability, widely used in Internet of Things (IoT) applications. Developed by Espressif Systems, it allows microcontrollers and devices to connect to Wi-Fi networks, making it ideal for smart systems and wireless communication projects.

The ESP8266 comes with a built-in microcontroller and can be programmed directly using the Arduino IDE, which makes it simple and beginner-friendly. It operates at **3.3V** and has **GPIO pins** that can interface with sensors, actuators, and displays. In IoT systems, the ESP8266 acts as the communication bridge between physical devices and cloud platforms, enabling real-time data monitoring and control. It is commonly used in applications such as home automation, smart appliances, weather stations, security systems, and remote monitoring devices. Its main advantages are low cost, compact size, integrated Wi-Fi, and ease of programming, while limitations include limited GPIO pins, lower processing power compared to ESP32, and power consumption concerns in battery- based applications.

Fig. 3.1.8 IOT (ESP8266)

3.2 SOFTWARE REQUIREMENTS

3.2.1 ARDUINO IDE

The Arduino IDE (Integrated Development Environment) is an open-source software application used to write, compile, and upload programs to Arduino boards and compatible microcontrollers. It provides a simple and user- friendly interface. The Arduino IDE supports with additional simplified functions to control hardware such as LEDs, sensors, motors, and displays. It features a text editor for writing code, a message area for errors and status updates, a console for output messages, and toolbars for compiling and uploading code. One of its key strengths is the availability of a wide range of libraries and examples, which simplify the integration of complex components like LCDs, Wi-Fi modules (ESP8266/ESP32), and IoT platforms.

Fig. 3.2.1 ARDUINO IDE

The IDE also supports serial monitoring, allowing users to debug and communicate with the microcontroller in real time. Being cross-platform, it runs on Windows, macOS, and Linux. Its advantages include ease of use, open- source nature, and a large community support, while its limitations include limited debugging features and lower optimization compared to professional IDEs. Despite this, Arduino IDE remains the most widely used tool for rapid prototyping and learning embedded systems.

3.2.2 Blynk software

Blynk is a popular IoT platform that allows developers to build mobile and web applications for controlling and monitoring hardware devices remotely. It provides a simple way to connect microcontrollers such as Arduino, ESP8266. and Raspberry Pi to the internet and control them using a smartphone app. Blynk works on a three- component architecture: the Blynk App, where the user creates a graphical interface with widgets to control devices; the Blynk Server, which manages communication between the app and hardware; and the Blynk Libraries, which enable the hardware to connect to the server. Using Blynk, users can easily implement features like switches, sliders, buttons, graphs, and notifications without writing complex code for mobile applications. It is widely used in home automation, smart agriculture, energy monitoring, and industrial IoT projects.

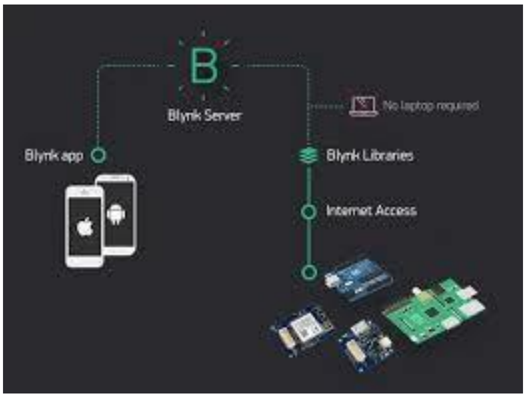

Fig. 3.2.2 BLYNK SOFTWARE

The platform supports both free and paid versions, with cloud hosting as well as the option to run a private local server for more control. Its main advantages include ease of use, cross-platform support, real-time monitoring, and minimal coding requirements, while limitations include dependency on internet connectivity and restrictions in the free version. Overall, Blynk has become one of the most beginner-friendly and powerful platforms for developing IoT-based applications.

4. LITERATURE OF REVIEW

1. Mansingh, P. B., Titus, T. J., & Devi, V. S. (2020, March), “A secured biometric voting system using RFID linked with the Aadhar database”, In 2020 6th International Conference on Advanced Computing and Communication Systems (ICACCS) (pp.1116-1119) IEEE.

   This paper proposes a secured biometric voting system integrating RFID with the Aadhar database. An active RFID tag replaces voter ID, and fingerprint verification ensures only genuine voters cast their votes. The system includes RFID readers, fingerprint scanners, and an LCD to display voter information. Alerts are triggered for fake votes, and voter data is sent to registered phone numbers. The integration with Aadhar provides high security and efficiency, preventing illegal voting or impersonation.

2. VH, P. K., & Rajput, R. P. (2020, July), “Design and implementation of convenient and compulsory voting system using finger print sensor and GSM technologies”, In 2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA) (pp. 446-450). IEEE.

   The study addresses vulnerabilities in India’s electronic voting system due to lack of voter identification. A fingerprint-based authentication system with GSM technology is proposed to prevent misuse of voting rights. Implemented using Arduino, the system collects and transfers data to local election offices for verification.This ensures authentication, quick processing, and reliable record keeping before elections. The approach enhances security and transparency in the voting process.

3. Rathee, G., Iqbal, R., Waqar, O., & Bashir, A. K. (2021), “On the design and implementation of a blockchain enabled e-voting application within iot-oriented smart cities.” IEEE Access, 9, 34165-34176.

   This paper focuses on a blockchain-enabled e-voting system for IoT-based smart cities. It addresses threats from malicious IoT devices using trust computation through a social optimizer. Blockchain ensures data immutability and prevents vote tampering. Security is analyzed against message alteration, DoS, DDoS attacks, and authentication delays. The proposed system provides a secure and transparent voting mechanism within smart city infrastructures.

4. Othman, A. A., Muhammed, E. A., Mujahid, H. K., Muhammed, H. A., & Mosleh, M. A. (2021, March), “Online voting system based on iot and ethereum blockchain.”, In 2021 international conference of technology, science and administration (ICTSA) (pp. 1-6). IEEE.

   An online voting system integrating IoT & Ethereum blockchain is presented to protect voter data from theft, eavesdropping, & tampering. Blockchain encryption safeguards votes & ensures integrity. The system is designed for both government & private organizations, streamlining elections & surveys. It eliminates traditional paper ballots, long queues, providing quick & efficient result.

5. Venkateshwara Rao.C.H., M. V. Pathi A., B. S. Sailesh A., (2022, January), "Arduino based electronic voting system with biometric and GSM features", In 2022 4th international conference on smart systems and inventive technology (ICSSIT) (pp. 685-688). IEEE.

   This study proposes an Arduino-based electronic voting system with fingerprint authentication and GSM communication. The system addresses vulnerabilities in conventional EVMs, such as tampering and double voting. Alerts are sent to registered mobile numbers when fake votes are detected. The integration of biometrics and GSM ensures reliability, security, fairness, and transparency in elections.

6. Vanitha, M., Sathyapriya, M., & Vishnuvardhan, P. (2022, October), "RFID based secure voting system with biometric authentication", In 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT) (pp. 1-6). IEEE.

   The paper introduces a multi-tier IoT-based voting system using RFID and biometrics. Two individuals handle the authentication process to start voting securely. GSM communication confirms vote casting to candidates. The design enhances security, eliminates fraud, and ensures the voting process is safe, functional, and transparent. The system aims to improve feasibility while addressing the limitations of traditional voting methods.

7. Chavan, S., Darekar, T., Chavhan, N., Deore, Y., & Suryawanshi, R. (2024, March), "Safe Count: A Two- Step Verification Based Voting System.", In 2024 International Conference on Emerging Smart Computing and Informatics (ESCI) (pp. 1-6). IEEE.

   This work presents "Safe Count," a voting system combining RFID and biometric verification. Facial recognition ensures each voter is unique, preventing proxy voting. The system cross-verifies RFID cards with biometric data to allow only eligible voters to cast ballots. It reduces election-related threats, ensures fairness, and increases accountability and transparency. The approach secures the electoral process effectively.

8. Chaithanya, D. J., Sinchana, S., Suprith, N. J., & Guru Kiran, R. (2024, May). "Campus Choice: Arduino Nano College Voting Machine", In 2024 1st International Conference on Communications and Computer Science (InCCCS) (pp. 1-5). IEEE.

   The paper develops a smart voting machine using Arduino Nano to overcome issues in traditional paper- based voting. The system is tamper-proof, ensures single voting per candidate, and provides immediate results. Instructions are displayed on an LCD for voter guidance. This design emphasizes fast, fair, and accessible vote counting, strengthening democratic processes using microcontroller-based technology.

9. Sidharth, P. V., Nagarajan, P. M., Johnson, D. G., Kumar, A. A., & Namitha, K. (2024, June), "Securing Democracy: The Two-Phase Authentication Approach to Electronic Voting." In 2024 3rd International Conference on Applied Artificial Intelligence and Computing (ICAAIC) (pp. 1788-1793). IEEE.
A two-phase authentication system combining facial recognition & fingerprint scanning is proposed for voting security. Arduino, GSM and Wi-Fi modules support voter identification. Invalid votes trigger alerts to election authorities. Biometrics enhance accuracy and accessibility, ensuring a secure voting process

10. Dixit, A., Gupta, A. K., Kaur, G., Jain, M., Pandey, R. K., & Sharma, A. (2024, December), " Enhancing Voting System Security and Accessibility through Biometric Authentication and IoT Integration.", In 2024 1st International Conference on Advances in Computing, Communication and Networking (ICAC2N) (pp. 1460- 1465). IEEE
This research integrates biometric authentication with IoT-enabled voting to enhance security and accessibility. Real-time verification occurs using IoT sensors capturing fingerprints, iris, or facial data. Multifactor authentication, including secure QR codes and voter ID cards, ensures privacy and accurate identification. The system is scalable and user-friendly, aiming to reduce impersonation, improve inclusivity, and establish trust in the voting process.

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5. DEFINITION OF THE PROBLEM & SCOPE OF THE RESEARCH WORK

5.1 Definition of the Problem

In the current voting systems, several challenges persist such as impersonation, multiple voting, booth capturing, and delays in result compilation. Traditional paper-based or even some electronic voting systems are vulnerable to human errors, data tampering, and lack of transparency. There is also limited real-time monitoring and verification of voter identity during the polling process.

To address these issues, there is a need for a secure, automated, and tamper-proof system that ensures only authenticated voters can cast their vote once, and that voting data is transmitted and tracked in real-time to maintain transparency and trust in the electoral process. The proposed system, **Secure Polling Booth Using RFID and Live Vote Tracking**, aims to solve these problems by integrating RFID technology for voter authentication and IoT-based live vote tracking for monitoring and result transparency.

5.2 Scope of the Research Work

The scope of this research work focuses on developing a **smart, IoT-enabled secure polling system** that combines **RFID-based voter identification** with **real-time vote monitoring** through a connected platform. The system will verify each voter using a unique RFID card, ensuring that only legitimate voters can participate. Once authenticated, the voter can cast their vote electronically, which will be instantly recorded and reflected on a live tracking interface for administrators or authorized officials. The project scope covers:

- Design and implementation of the RFID-based authentication module.
- Integration of Arduino/ESP8266 with a database and cloud server for live vote tracking.
- Development of a secure interface for viewing real-time voting statistics.
- Ensuring data integrity and preventing duplicate or unauthorized voting.

6. ARCHITECHTURAL DESIGN

6.1 Conceptual Design

The proposed **secure electronic voting system** is carefully designed to ensure **security,**

transparency, and efficiency by integrating several advanced technologies such as **RFID**, **password-based authentication**, **GSM communication**, and **IoT-based monitoring**. This multi-layered design enhances both the integrity and reliability of the voting process while reducing the possibility of fraud or tampering.

6.1.1 Two-Factor Authentication

- **RFID Card:**

In this system, each registered voter is provided with a **unique RFID card**, which functions as a **personal digital identification token**. When the voter places the RFID card near the reader, the **RFID reader scans the unique ID** encoded on the card and transmits it to the **Arduino microcontroller**. The Arduino then verifies this ID against the **stored database of authorized voters**. Only if the ID matches a valid entry does the system allow the voter to proceed further.

- **Password Verification:**

After the RFID verification is successfully completed, the system moves to the **second layer of authentication**—password verification. The **LCD screen** displays a message prompting the voter to **enter their unique password** using the **3×4 matrix keypad**. The entered password is then cross- checked with the password stored in the system database for that particular voter. If both the RFID and password credentials are verified successfully, the voter is then **granted access to the voting stage**.

This **two-step authentication mechanism** greatly enhances security, effectively preventing **impersonation, multiple voting attempts**, and **unauthorized access** to the voting system. Only genuine and verified voters are permitted to cast their votes, ensuring the integrity of the election process.

- **Vote Casting & Feedback**

Once the voter's identity is confirmed through the two-factor authentication, they can now proceed to **cast their vote**. The system interface consists of **push buttons**, each corresponding to a specific candidate or party. The voter simply **presses the button** for the candidate of their choice.

During this process, the **LCD display (16×2)** provides **clear visual feedback**, showing the voter's name, the voting instructions, and the **selected candidate's name or symbol**. This real-time feedback helps ensure that the voter is aware of their selection and can confirm the accuracy of their choice before final submission. This step enhances the **user- friendliness and transparency** of the system by keeping the voter informed throughout the process.

- **Transparency & Real-Time Monitoring**

To further enhance **trust and accountability**, the system employs **GSM and IoT-based communication technologies**. Once a vote is successfully cast, the **GSM module (SIM800L)** automatically sends an **SMS confirmation** to the voter's registered mobile number. This message includes essential details such as the **confirmation of the vote** and serves as a receipt that the voter's choice has been recorded.

Simultaneously, the **IoT module (ESP8266 NodeMCU)** transmits the **updated vote count data** to a **cloud platform** in real time. Election authorities can monitor this live data from a **remote dashboard**, ensuring continuous supervision of the voting process. This real-time Monitoring eliminates the need for manual counting, reduces the chances of human error, and helps detect any irregularities immediately.

This combination of **GSM-based feedback** and **IoT-based transparency** creates a **secure, tamper-resistant, and highly accountable voting system**, ensuring that every vote is properly counted and recorded.

6.2 Implementation

6.2.1 Hardware Architecture

The system is built upon a well-structured hardware framework centered around the **Arduino Uno microcontroller**, which acts as the main processing unit responsible for managing all operations including RFID scanning, password validation, input recognition, and communication with other modules.

- **Arduino Uno:** Serves as the **central controller** that coordinates every function wise within the system — from authentication and vote recording to message transfer and IoT connectivity.
- **RFID Reader (RDM6300):** Reads the **unique ID** from the voter's RFID card and transmits it to the Arduino for authentication.
- **RFID Cards:** Assigned individually to voters, functioning as **digital ID cards** that uniquely identify each person in the system.
- **3×4 Matrix Keypad:** Enables the voter to **enter their password** securely during the second phase of authentication.
- **Push Buttons:** Represent the **candidates or parties**. Each button press corresponds to a vote for a specific candidate.
- **LCD Display (16×2):** Provides real-time **visual feedback** such as voter identification confirmation, voting instructions, and the name of the selected candidate.
- **GSM Module (SIM800L):** Sends **SMS notifications** to voters confirming successful voting and can alert authorities in case of irregularities.
- **IoT Module (ESP8266 NodeMCU):** Uploads live **vote counts and statistics** to an online server or cloud platform for **remote supervision**.
- **Power Supply (5V regulated):** Provides a **stable and consistent voltage** to all connected components, ensuring smooth system operation.This combination of hardware elements ensures the system is **robust, reliable, and secure**, supporting smooth interaction between all modules.

6.2.2 Software / Working Flow

The **Arduino program** acts as the system's brain and is programmed to manage the entire voting process in a systematic sequence.

- **Step 1: RFID Verification**
  The system begins by initializing all hardware modules. When a voter brings their RFID card near the reader, the card's **unique identification number** is captured and compared against the system's **authorized database**. Only recognized IDs are permitted to proceed.
- **Step 2: Password Verification**
  If the RFID ID matches, the **LCD screen prompts** the voter to input their **password** using the keypad. The entered password is verified against the stored one in memory. If correct, the voter can access the voting stage; otherwise, the system denies access and resets for the next user.
- **Step 3: Vote Casting**
  Once authenticated, the voter presses the **button** assigned to their preferred candidate. The **Arduino** detects the input and increments the vote count for that candidate.
- **Step 4: Feedback and Transmission**
  Immediately after casting the vote, the **LCD display** shows confirmation of the voter's selection. The **GSM module** sends an SMS confirmation to the voter's phone, while the **IoT module** updates the cloud database with the revised vote totals.
- **Step 5: Continuous Monitoring**
  Election authorities can view and monitor **real-time voting data** through the dashboard.

  This ensures continuous transparency &instant detection of any voting discrepancies.

This **multi-layered process**, combining **RFID and password-based augmented authentication** with **GSM and IoT-based communication**, forms a **highly secure and transparent electronic voting system**. It addresses the limitations of traditional EVMs and manual voting systems by providing **accuracy and real-time visibility Security and accountability** throughout the election.

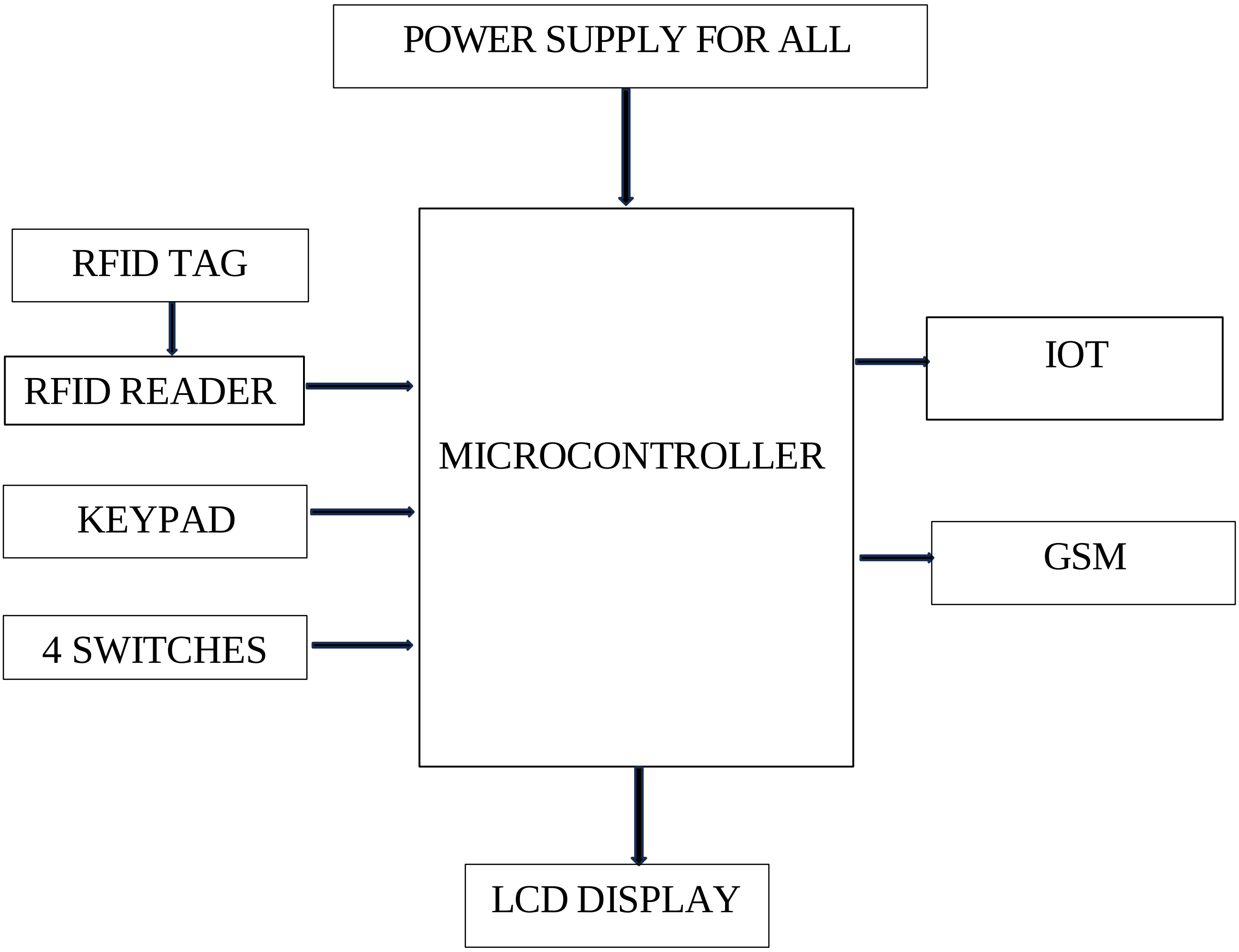

Fig. 6.2 BLOCK DIAGRAM

7. RESULTS & DISCUSSION

7.1 RESULTS

Our RFID-based voting system worked well during testing. Each voter received an RFID card and a personal password. When the card was scanned, the system quickly verified the voter's identity,displayed the voter's name on the LCD screen, and requested the password. Only after entering the correct password could the voter select a candidate using the push buttons. The selected nameappeared immediately on the LCD. At the same time, the vote count updated automatically on the IoT platform so election staff could see results in real-time. A text message was sent to the voter's phone to confirm their choice. This setup prevented duplicate votes, reduced manual errors, and provided instant feedback. The prototype operated smoothly in repeated tests and proved to be secure, fast, and easy to use compared to traditional paper ballots or basic electronic machines.

7.2 DISCUSSION

This project shows how combining simple technologies can create a more reliable voting experience.By using RFID cards with passwords, we prevented people from voting multiple times orimpersonating others. The live IoT updates allowed officials to track the numbers as they came in, making it harder for anything suspicious to go unnoticed. The SMS confirmation gave voters confidence that their vote was recorded accurately.It also improved the overall process. Since the machine counted votes automatically, there was noneed for lengthy manual tallies at the end of the day. The clear LCD instructions and straightforward push buttons meant that even first-time users had no difficulty casting a vote.

8. CONCLUSION & FUTURE ENHANCEMENT

8.1 Conclusion

The proposed **Secure Polling Booth using RFID and Live Vote Tracking** system successfully demonstrates a **modern, reliable, and transparent voting mechanism** that addresses many of the challenges faced in traditional voting systems. By integrating **RFID-based voter identification** with **password verification**, the system ensures that only **authorized voters** can cast their votes, thereby preventing impersonation and multiple voting attempts. The inclusion of **GSM communication** and **IoT-based real-time monitoring** adds an extra layer of **transparency and trust**, as every vote is immediately recorded, confirmed, and updated on a cloud platform accessible to election authorities. Through the combination of these technologies, the system achieves **accuracy, security, and efficiency** in the voting process while significantly reducing manual effort and human error. It provides a digital transformation to conventional election methods, ensuring a **tamper- proof, verifiable, and accessible** voting experience for both voters and administrators. Overall, this project proves to be a **promising step toward smart and transparent electronic voting systems**, suitable for deployment in schools, institutions, or even large-scale elections with further optimization.

8.2 Future Enhancement

In the future, this project can be further enhanced and expanded to improve its **scalability, usability, and security**. Some possible advancements include:

1.Blockchain Integration:

Using blockchain technology to store and verify votes can create an immutable and tamper-proof ledger, ensuring complete transparency and traceability of election data**.**

2.Enhanced Cloud Dashboard:

Developing a **dedicated web or mobile dashboard** for election authorities and the public can allow **real-time visualization of voting statistics**, turnout rates, and system health monitoring

3.Offline Data Backup:

Adding a feature for this ensures the voting process continues securely even if internet connectivity fails temporarily.

4. **Solar-Powered Operation:**

   Implementing **solar energy modules** can make the system **self- sustainable and portable**, allowing its use in remote or rural areas without stable electricity.

5. **Multilingual and Voice-Assisted Interface:**

   Incorporating a **multilingual or voice-guided user interface** can improve accessibility for **elderly or differently-abled voters**, ensuring inclusivity.

By implementing these future enhancements, the system can evolve into a **comprehensive, next- generation e-voting platform** capable of supporting **national-level elections** with high efficiency, trust, and technological resilience.

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